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Vortex nanolaser based on a photonic disclination cavity

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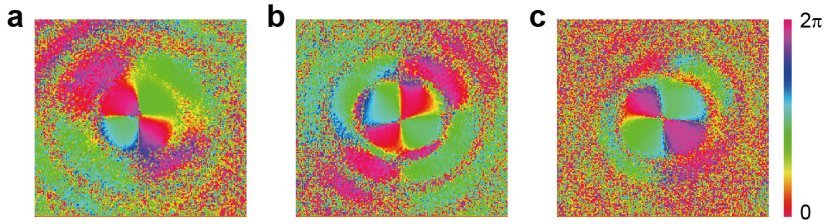
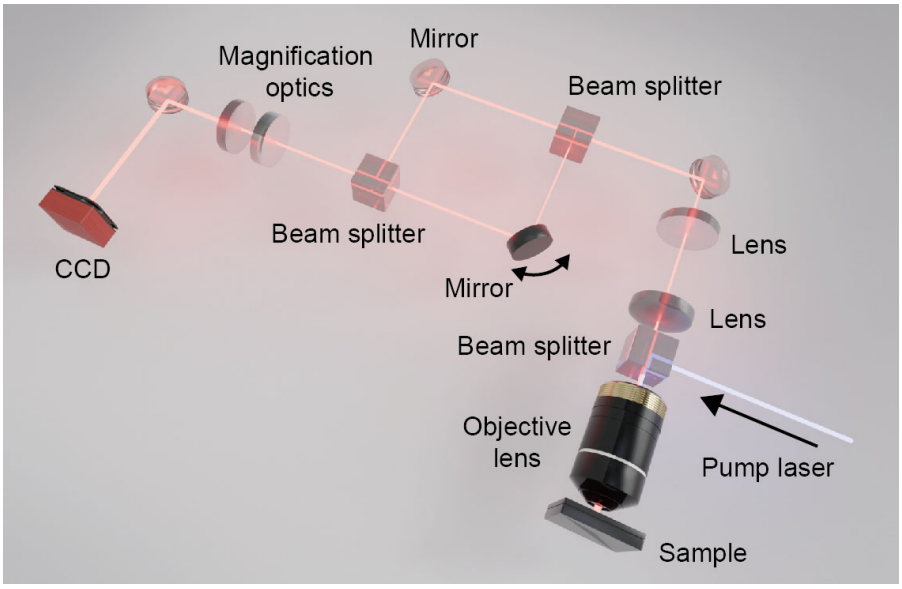


Supplementary Fig. 1. Intra-modal phase. The intra-modal phases, arctan (S_3/S_2), are calculated using the measured Stokes parameters of Figs. 5a (**a**), 5b (**b**), and 5c (**c**).



Supplementary Fig. 2. Off-center self-interference measurement setup. To investigate the vortex properties of photonic disclination cavities, off-center self-interference pattern measurements were performed. Two mode profiles split by beam splitters were overlapped by adjusting mirrors, leading to modifications in both the beam path and central positioning. Then, the self-interference of two doughnut-shaped beams split from the same lasing emission generated interference patterns with two fork-shaped fringes[7].

	Threshold peak power (mW)	Threshold power density (kW/cm^2)	Cavity size (μm x μm x μm)	Q factor	Cavity type	Optical pumping condition
This work	**0.08**	**1.1**	**3.0 x 3.0 x 0.29**	**2 x 10^4**	**Disclination**	**Pulsed (1 MHz, 20 ns) at room temp**
Science 353, 464 (2016) (Ref. [3] of the manuscript)	-	100	9 x 9 x 1.5	-	Microring	Pulsed (50 kHz, 8 ns) at room temp
Science 368, 760 (2020) (Ref. [4])	-	26	7 x 7 x 0.2	-	Microring	Pulsed (50 kHz, 8 ns) at room temp
Nat. Commun. 11, 4862 (2020) (Ref. [5])	1.5 x 10^5	1.9 x 10^5	10 x 10 x 0.6	6 x 10^3	Spiral grating	Pulsed (1 kHz, 100 fs) at room temp
Nat. Photon. 13, 283 (2019) (Ref. [6])	0.94	0.3	8 x 7 x 7	-	DBR	CW at low temp (4 K)
Science 367, 1018 (2020) (Ref. [7])	5.3 x 10^5	4.2 x 10^4	40 x 40 x 0.22	10^8 (infinite array)	BIC	Pulsed (1 kHz, 100 fs) at room temp
Laser & Photon. Rev. 16, 2100574 (2022) (Ref. [8])	2.5 x 10^9	3.2 x 10^7	100 x 100 x 0.17	10^4	BIC	Pulsed (1 kHz, 120 fs) at room temp
Phys. Rev. Lett. 125, 013903 (2020) (Ref. [16])	-	120	13 x 8 x 0.2	-	Topological edge	Pulsed (12 kHz, 5 ns) at room temp

Supplementary Table 1. Comparison of our photonic disclination nanolaser with other types of vortex nano-/microlasers.

Supplementary Note

1. Analysis of photonic crystal slab modes

Maxwell's equations in an isotropic non-magnetic optical medium read

$$-i\nabla \times \boldsymbol{E} = \omega\mu_0\boldsymbol{H},\ \ i\nabla \times \boldsymbol{H} = \omega\varepsilon\boldsymbol{E},\ \ \nabla \cdot (\varepsilon\boldsymbol{E}) = \nabla \cdot \boldsymbol{H} = 0, \tag{1}$$

where μ_0 is the permeability of free space and $\varepsilon = \varepsilon(\boldsymbol{r})$ is the (position $\boldsymbol{r}$-dependent) permittivity. Consider a quasi-two-dimensional photonic crystal slab in which the z-dependence of ε, $\boldsymbol{H}$, and $\boldsymbol{E}$ can be neglected. Expanding the curl term,

$$\left(-i\,\partial_y, i\,\partial_x\right)E_z = \omega\mu_0\left(H_x, H_y\right), \quad -i(\nabla \times \boldsymbol{E})_z = \omega\mu_0 H_z. \tag{2}$$

$$\left(i\,\partial_y, -i\,\partial_x\right)H_z = \varepsilon\omega\left(E_x, E_y\right), \quad i(\nabla \times \boldsymbol{H})_z = \varepsilon\omega E_z. \tag{3}$$

Thus, the slab modes decouple into transverse-magnetic (TM) modes with nonzero (H_x, H_y, E_z), and transverse-electric (TE) modes with nonzero (E_x, E_y, H_z).

Consider TE modes described by the profile $H_z(x, y) = H_z(\boldsymbol{r}_\perp)$ satisfying

$$\omega^2\mu_0 H_z = -\nabla_\perp \cdot \left(\frac{1}{\varepsilon}\nabla_\perp H_z\right) = -\frac{1}{\varepsilon}[\nabla_\perp^2 H_z - \nabla_\perp(\ln\varepsilon) \cdot (\nabla_\perp H_z)]. \tag{4}$$

Let us assume the structure has a point group symmetry about the origin and expand H_z as a multipole series in cylindrical coordinates $\boldsymbol{r}_\perp = (r\cos\varphi, r\sin\varphi)$,

$$H_z(\boldsymbol{r}_\perp) = \sum_m f_m(r)e^{im\varphi}, \tag{5}$$

or, in Fourier space,

$$H_z(\boldsymbol{k}_\perp) = 2\pi\sum_m i^{-m}F_m(k)e^{im\varphi}, \tag{6}$$

where

$$F_m(k) = \int_0^\infty f_m(r)J_m(kr)r\,dr. \tag{7}$$

Thanks to the C_n point group symmetry, the modes are invariant (up to a phase factor) under a rotation by $2\pi/n$, i.e. $H_z(r,\varphi+2\pi/n)=H_z(r,\varphi)e^{i2\pi l/n}$ for some integer l. Without loss of generality, we can take $|l|\leq n/2$, where l is the angular momentum. Employing the multipole expansion, rotational invariance requires all m with nonzero coefficients f_m to satisfy $2\pi(m-l)/n=0 \bmod 2\pi$. This can be satisfied if $m=l+qn$, for any integer q. Thus, modes with angular momenta l will have nonzero multipole expansion coefficients f_l, $f_{l\pm n}$, $f_{l\pm 2n}$, and so on.

To understand how these expansion coefficients relate to the emitted vector beams, we can rewrite the Maxwell's curl equations in cylindrical coordinates as follows:

$$\boldsymbol{E}=\frac{1}{r}\partial_\varphi H_z\hat{r}-\partial_r H_z\hat{\varphi}. \tag{8}$$

Making use of $\hat{r}=\cos\varphi\hat{x}+\sin\varphi\hat{y}$, $\hat{\varphi}=-\sin\varphi\hat{x}+\cos\varphi\hat{y}$, $\hat{\mathbf{e}}_{R,L}=\frac{1}{\sqrt{2}}(\hat{x}\mp i\hat{y})$, and collecting terms, we obtain the modal electric field profile in the circular polarization basis as follows:

$$\boldsymbol{E}=\frac{\partial_\varphi H_z}{r\sqrt{2}}\left(e^{i\varphi}\hat{\mathbf{e}}_R+e^{-i\varphi}\hat{\mathbf{e}}_L\right)-i\frac{\partial_r H_z}{\sqrt{2}}\left(e^{i\varphi}\hat{\mathbf{e}}_R-e^{-i\varphi}\hat{\mathbf{e}}_L\right), \tag{9}$$

which is decomposed into TE-like (first term) and TM-like (second term) components. Alternatively, we can decompose in terms of the circular polarization components as follows:

$$\boldsymbol{E}=\frac{1}{\sqrt{2}}\{[\frac{1}{r}\partial_\varphi H_z+i\partial_r H_z]e^{-i\varphi}\hat{\mathbf{e}}_L+[\frac{1}{r}\partial_\varphi H_z-i\partial_r H_z]e^{i\varphi}\hat{\mathbf{e}}_R\}. \tag{10}$$

The polarization azimuth θ satisfies $|\sigma|e^{i\theta/2}=E_R^*E_L$ (ref. S1). For real H_z, introducing $\phi=\mathrm{Arg}\left[\left(\frac{1}{r}\partial_\varphi H_z+i\,\partial_r H_z\right)e^{-i\varphi}\right]$ simplifies to $\theta/2 \bmod 2\pi=2\phi$.

We can now relate the lasing mode profiles to the H_z symmetry eigenvalue l. The nonmodified and modified disclination boundaries with expanded core host defect modes with $l=0$. In the continuum limit (weakly confined modes), we can approximate H_z using the lowest order ($m=0$)

term in the multipole expansion; thus $\partial_\varphi H_z \approx 0$ and we are left with a TM-like electric field profile, corresponding to a polarization vortex:

$$\boldsymbol{E}_{l=0} = -i\frac{1}{\sqrt{2}}f_0'(r)\left(e^{i\varphi}\hat{\mathbf{e}}_R - e^{-i\varphi}\hat{\mathbf{e}}_L\right). \tag{11}$$

On the other hand, the modified disclination boundaries with shrunken core supports defect modes with nonzero l. The defect modes are described by a purely real H_z profile, corresponding to equal weight superpositions of $+l$ and $-l$, i.e. $H_z^{(+)} \propto f_{|l|}(r)\cos(l\varphi)$ or $H_z^{(-)} f_{|l|}(r)\sin(l\varphi)$. This will lead to electric field profiles of the form

$$\begin{aligned}\boldsymbol{E}_+ = \frac{1}{\sqrt{2}}\{&[-\frac{lf(r)}{r}\sin(l\varphi) + if'(r)\cos(l\varphi)]e^{-i\varphi}\hat{\mathbf{e}}_L \\ &+[-\frac{lf(r)}{r}\sin(l\varphi) - if'(r)\cos(l\varphi)]e^{i\varphi}\hat{\mathbf{e}}_R\}.\end{aligned} \tag{12}$$

$$\begin{aligned}\boldsymbol{E}_- = \frac{1}{\sqrt{2}}\{&[\frac{lf(r)}{r}\cos(l\varphi) + if'(r)\sin(l\varphi)]e^{-i\varphi}\hat{\mathbf{e}}_L \\ &+[\frac{lf(r)}{r}\cos(l\varphi) - if'(r)\sin(l\varphi)]e^{i\varphi}\hat{\mathbf{e}}_R\}.\end{aligned} \tag{13}$$

For small r, $f_{|l|}(r) \propto r^{|l|}$ and thus these expressions further simplify to

$$\boldsymbol{E}_+ = \frac{ir^{|l|-1}}{\sqrt{2}}\{e^{i(|l|-1)\varphi}\hat{\mathbf{e}}_L - e^{-i(|l|-1)\varphi}\hat{\mathbf{e}}_R\}. \tag{14}$$

$$\boldsymbol{E}_- = \frac{r^{|l|-1}}{\sqrt{2}}\{e^{i(|l|-1)\varphi}\hat{\mathbf{e}}_L + e^{-i(|l|-1)\varphi}\hat{\mathbf{e}}_R\}. \tag{15}$$

Setting $|l| = 2$ will reproduce the structure of the hybrid modes $HE^{o,e}$.

Finally, we need to explain the difference between the in-slab and emitted field profiles. The simulated H_z slab mode profiles show mainly a three-fold rotational symmetry, corresponding to $|l| = 3$. Such higher-order multipole modes are, however, dark and will weakly couple to the out-of-plane radiation. The C_5 point group symmetry of the structure will result in coupling between states with angular momenta l and $l \pm 5$. Thus, it is sufficient to take $|l| \leq 2$ without loss of

generality. In other words, the $l = \pm 3$ multipole will also couple to the $l = \mp 2$ quadrupole mode. Since this is a lower-order multipole, its r-localized field profile will couple more strongly to the out-of-plane radiation, explaining why we see the $|l| = 2$ symmetry in the output beam. Another way of understanding this is by looking at the corresponding H_z field profiles: as r increases for a fixed φ, the sign of H_z change. Thus, the mode field profile resembles a ring of (radiating) dipoles with φ-dependent orientation. The radial localization of the defect modes is an essential component here. If we consider delocalized Bloch waves formed from quadrupole modes, we find that they are non-radiative[S2].

2. Energy tunability of disclination states

Topological in-gap modes have been observed in a number of recent experiments in photonics and condensed matter. Even in the recent realization of the photonic SSH model, a canonical model that describes a one-dimensional topological insulator, topological in-gap modes were observed in the absence of chiral symmetry[S3,S4]. Without chiral symmetry, the in-gap corner states are not necessarily pinned at the center of the bulk energy gap. However, the energy of the in-gap modes is generally tunable within the bulk gap as demonstrated below, and thus, in-gap modes of topological origin can be experimentally achieved in photonic crystals. Similar situations also occur in 2D higher-order topological insulators (HOTIs) that exhibit the characteristics of zero-dimensional corner states in the gap[S5-S7].

One might wonder if in-gap modes shifted away from the gap center lose their topological characteristics. However, even when the in-gap modes are completely buried in the bulk continuum, the bulk band topology manifests in the form of the *filling anomaly* as long as the relevant symmetry is preserved[S8]. Specifically, the topological and non-topological bulk states can

be distinguished by counting the number of occupied states, regardless of the energy position of the in-gap modes, and the topological bulk hosts fractional charges at the corners. For instance, in C_n symmetric HOTIs, filling anomaly can result in the formation of corner states. More explicitly, when the system has open boundaries, the higher-order topological phase exhibits filling anomaly, which is protected by C_4 symmetry and time-reversal symmetry for each band, and the set of rotational topological invariants guarantees the emergence of fractional corner charges[S9,S10].

In the case of the disclination defects in crystals as in our work, the presence of fractional charges can be analyzed using rotational symmetry[S11,S12]. Despite the absence of chiral symmetry, recent discoveries have revealed that the presence of quantized fractional disclination charges in C_n symmetric 2D topological crystalline insulators (TCIs) is related to the general interpretation of their topological properties.

Next, to show the consistency of our work with previous literature, we revisit the disclination model with NN and NNN couplings (Fig. R1). Because of the presence of C_{4v} symmetry in the lattice, two topological phases can be distinguished based on the ratio of inter-cellular (t_1) and intra-cellular (t_2) hopping. In addition to t_1 and t_2, our disclination model is governed by NNN coupling (t_{nn}), which disrupts the corner modes to pin the zero-energy level. While the total Hamiltonian exhibits broken chiral symmetry, it still respects C_{4v} symmetry in our geometry. Figure R1 shows how broken chiral symmetry and the local boundary effect change the band structure, causing disclination states to appear inside the bandgap. NNN interactions break chiral symmetry and make the spectrum asymmetric; however, they do not eliminate edge, corner, and disclination topological states. Furthermore, the TB simulation of Fig. R1 shows that the energy of disclination modes can be shifted within the bulk gap, depending on t_2/t_1. Such energy tunability

of topological disclination states increases the feasibility of achieving the topological in-gap modes despite the lack of chiral symmetry in the system.

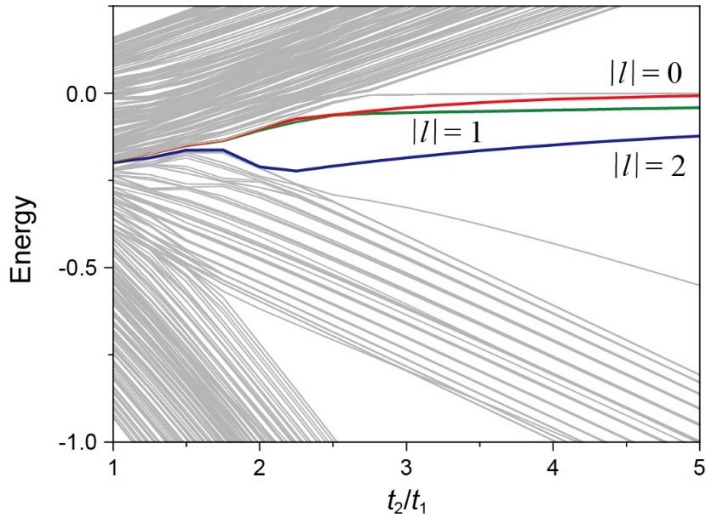


Fig. R1. Calculated in-gap states ($|l|$ = 0, 1, and 2) in the C_5 disclination cavity of Fig. 2d as a function of t_2/t_1. t_2/t_1 was 5.0 in Fig. 2d.

Furthermore, we modify the disclination boundary in the C_5 disclination cavity (Fig. R2a). The energies of in-gap states are calculated by varying the boundary hopping parameter, t_w' (Fig. R2b). Other hopping parameters are fixed. This simulation result indicates that well-confined in-gap states with degeneracy occur at t_w' = -0.05. With increasing $|t_w'|$, the degenerate disclination states are split into two degenerate modes and one nondegenerate mode, shifted to the conduction band or valence band, and eventually immersed in the bulk continuum. If the disclination design fails to satisfy the topological feature, there will be trivial defect states that originate only from the top or bottom bands. We emphasize that this tuning is straightforward in experiment. For example, increasing the air holes in the disclination boundary can satisfy the increase in $|t_w'|$.

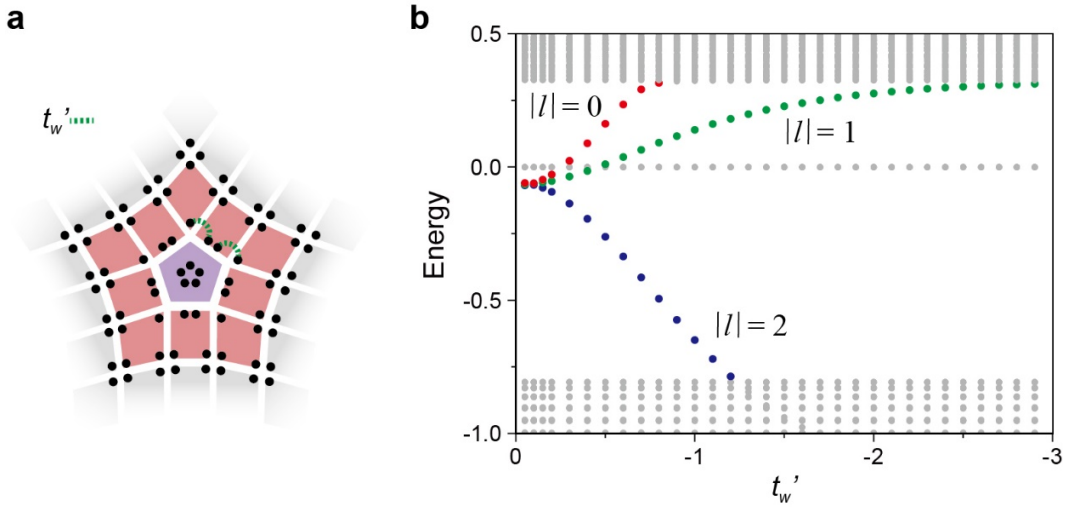


Fig. R2. a, C_5 disclination cavity with a boundary hopping parameter, t_w' (green dashed line). **b,** Calculated in-gap states ($|l|$ = 0, 1, and 2) in the cavity of **a** as a function of t_w'.

To sum up, despite the absence of chiral symmetry in our system, we show that disclination states can be achieved in the bulk gap, and their energies are even tunable by modifying the model parameters. In addition, the topological characteristics of the HOTI manifests in the form of filling anomaly, irrespective of the energy of in-gap modes, leading to fractional disclination charges. We believe that this concept is very useful for designing a new topological disclination cavity in which vortex and anti-vortex modes are excited with wavelength-scale mode volumes.

Supplementary References

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